



TARGET-90

JKBOSE CLASS-11TH

THE LIVING WORLD

One shot

BIOLOGY

ARSHID Sir



Topics to be Covered

1

Living world (MIC)



Most important Concepts...

2



Biodiversity

≡ Biodiversity

- "Bio" — means "life"
- "diversity" — means "forms / variety"

≡ Biodiversity — refers to number and types of living organisms + nt on planet Earth.

≡ 1.7 to 1.8 Million Sps. — Scientifically named and classified } Taxonomically Known Sps.

- 1.2 million — include Animal Sps.
- 0.5 million — include Plant Sps.

≡ Among Animals — largest group is Arthropoda (insects) — > 10,25,000 Sps.

≡ Among Plants — largest group is Angiosperms (Flowering plants) — > 80%.



Need For Classification

≡ There are millions of plant and animal sps.



Varying greatly in their form, structure and complexity.

≡ It is very difficult to study all of them individually.

≡ In order to make study easy — organisms have been divided into different categories/ranks on the basis of similarities and differences.



Taxonomy



- ≡ Taxonomy — derived from two Greek words $\begin{cases} \text{"Taxis"} - \text{Arrangement.} \\ \text{"Nomos"} - \text{laws / rules.} \end{cases}$
- ≡ Taxonomy — branch of science which deals with process of classification of living organisms by following rules/laws.
- ≡ Term taxonomy — given by "A.P deCandolle" in his book — "Theory of Elementary Botany"
- ≡ Carolus Linnaeus — regarded as "Father of taxonomy."



Taxonomy Involves Following Steps

a. CHARACTERISATION :-

≡ Organism is described according to its all morphological, anatomical, physiological and behavioural properties.

b. Identification :-

≡ organism is recognised from other already known organisms and is assigned to a particular taxonomic group.

c. Classification :-

≡ Organism is grouped into convenient categories on the basis of some easily observable characters.

d. Nomenclature :-

≡ Naming of an organism according to established rules.



Systematics

- Word Systematics is derived from latin word — "Systema"
↓
Which means systematic arrangement of living organisms.
 - Term "Systematics" — used by Carolous Linneaus in his book "Systema Naturae"
 - Systematics deals with
 - diversity of organisms. (Taxonomy)
 - AND
 - Evolutionary relationship b/w organisms (PHYLOGENY).
 - In modern times, Systematics includes
 - TAXONOMY
 - AND
 - PHYLOGENY (Evolutionary relationship b/w organisms).
- Systematics = Taxonomy + Phylogeny



Nomenclature



- ≡ PROCESS of assigning a name to an organism.
- ≡ Two types of Names have been given to organisms.

a. Common / Vernacular name / local name / regional names:

- ≡ These names are given to organisms in a particular language & region of world.
- ≡ These names are full of ambiguity. (confusion)

b. Scientific names / technical names:

- ≡ These names are given by biologists and is understood to represent a particular organism in every part of world.
- ≡ No ambiguity.



Binomial Nomenclature

imp.



≡ Given by Carolus Linnaeus.

≡ The system of providing a name with two components is k/d Binomial nomenclature.

→ Genus / Generic name.
AND
→ Specific Epithet.

Ex. Solanum tuberosum (potato)
↓ ↓
Genus / Generic name Specific Epithet

*

Note: Species name is the complete name of a species (Genus + Specific Epithet).

≡ Carolus Linnaeus — used binomial nomenclature in his book "Species plantarum."



Rules of Binomial Nomenclature

imp.

- ⚡ Biological names are generally taken from latin language irrespective of their origin (or are latinized).
- ⚡ Biological names consists of two components.
 < Genus / Generic name
 < Specific epithet.
- ⚡ Scientific names are printed in italics or underlined separately when handwritten to indicate their latin origin. Ex. Solanum tuberosum.
- ⚡ First letter of genus starts with capital letter while the specific epithet starts with small letter.
- ⚡ Name of author/discoverer is written after specific epithet in abbreviated form.
It is not underlined nor italisized. Ex. Mangifera indica (Linn.) → C. Linneaus (Author)



Codes of Nomenclature

- **Agreed convention for naming biological groups.**
 - ⚡ ICBN – International Code for Botanical Nomenclature. (For plants)
 - ⚡ ICZN – International Code of Zoological Nomenclature. (For animals)

Common Names	Scientific Names
1. <u>POTATO</u>	<u>Solanum</u> <u>tuberosum</u>
2. <u>Brinjal</u>	<u>Solanum</u> <u>melongena</u>
3. <u>Makoi / Black night shade</u>	<u>Solanum</u> <u>nigrum</u>
4. <u>Lion</u>	<u>Panthera</u> <u>leo</u>
5. <u>LEOPARD</u>	<u>Panthera</u> <u>pardus</u>
6. <u>Tiger</u>	<u>Panthera</u> <u>tigris</u>
7. <u>Dog</u>	<u>Canis</u> <u>familiaris</u>
8. <u>wolf</u>	<u>Canis</u> <u>lupus</u>
9. <u>Cat</u>	<u>Felis</u> <u>catus</u>
10. <u>Man</u>	<u>Homo</u> <u>sapiens</u>
11. <u>Housefly</u>	<u>Musca</u> <u>domestica</u>
12. <u>Mango</u>	<u>Mangifera</u> <u>domestica</u>
13. <u>wheat</u>	<u>Triticum</u> <u>aestivum</u>

Home work



Taxonomic Categories

- ≡ Classification — is not a single step process.
- ≡ It is multistep process and thus involves hierarchy of steps in which each step represents a rank / category / Taxon. (pl. taxa).

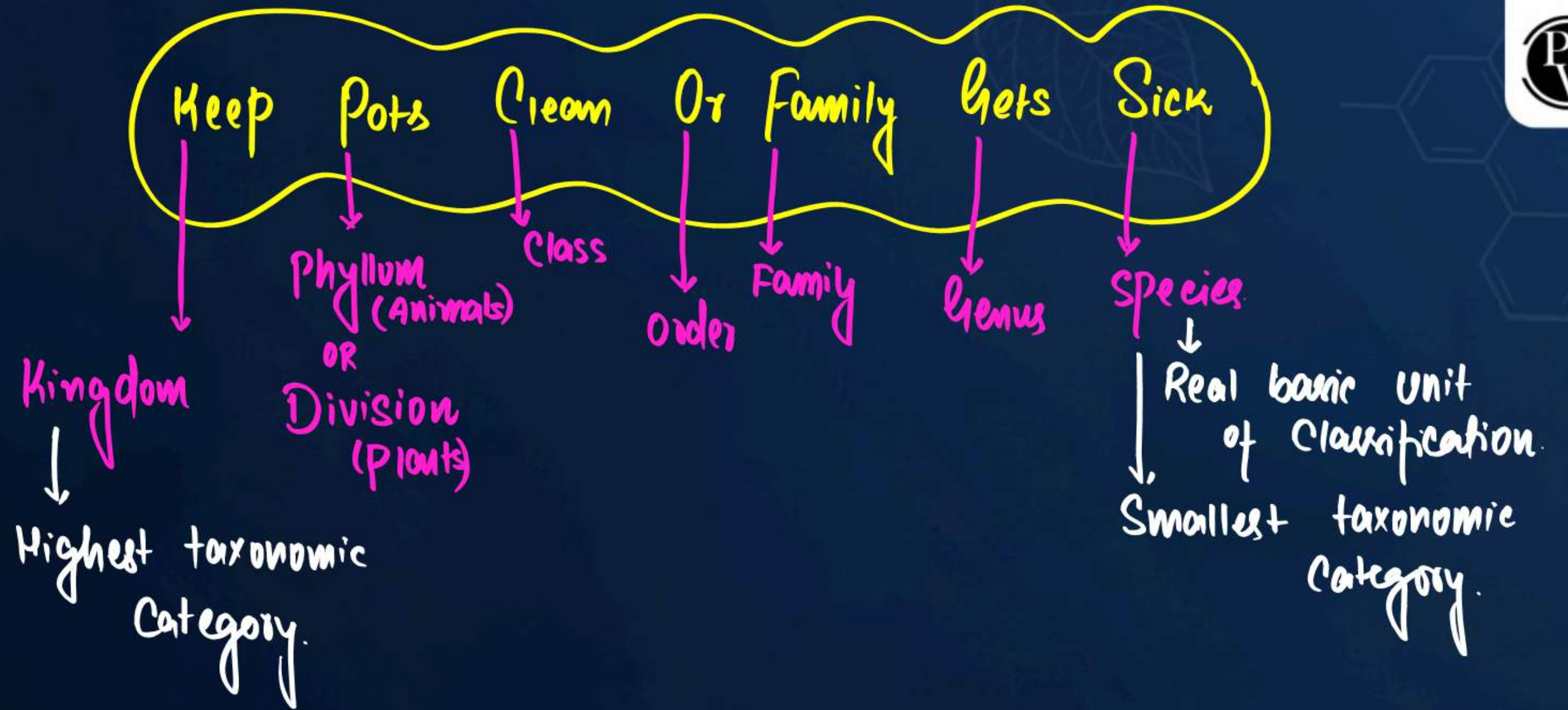
* Taxon — unit of classification.

NOTE:- Thin line of difference b/w Taxon and Category.

Taxon - Singular
Taxa - plural

<u>TAXON</u>	<u>CATEGORY</u>
→ It represents living entity.	→ It represents rank.
→ Ex: <u>Panthera</u> , <u>Solanum</u>	→ <u>Genus</u>

* There are 07 main taxonomic categories which are obligate / essential i.e., why they are strictly used at the time of any plant classification.



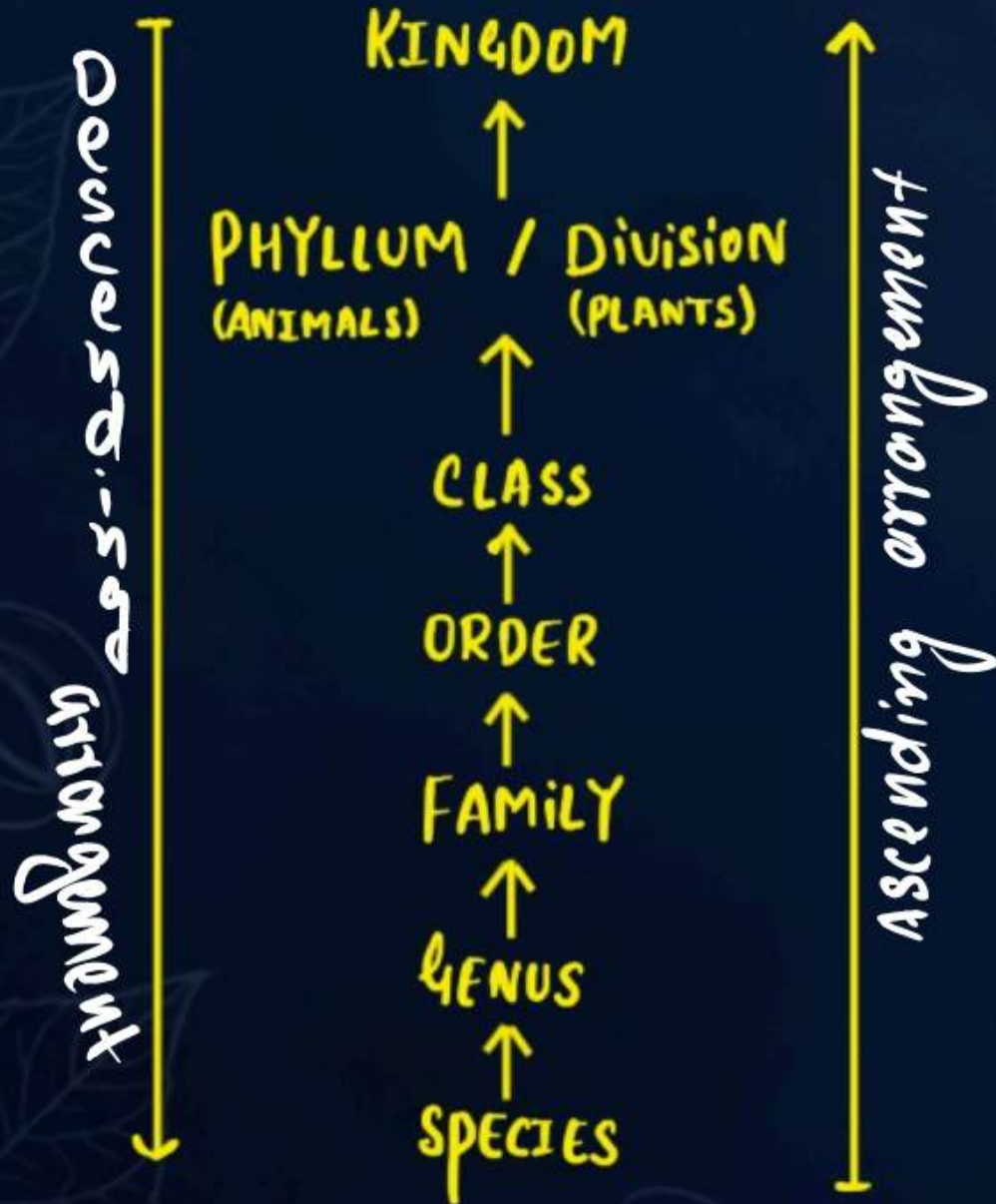


Taxonomic Hierarchy

≡ Arrangement of taxonomic categories — either in ascending or descending order.

NOTE: AS WE MOVE FROM SPECIES TO KINGDOM:

- ①. No. of similar characters i.e., specificity — Decreases.
- ②. No. of general characters — Increases.
- ③. Species diversity — Increases
- ④. Complexity of classification — Increases.





Species

imp.



≡ Real basic unit of classification.

* NOTE: → Unit of classification — TAXON.

≡ Species — group of individual organisms with fundamental similarities as a species.

↓
defined by "Biological Species Concept" — proposed by "Ernst Mayr"

As per this Concept, Species is a group of organisms which:

a. are closely related (Structurally and functionally)

c. Can interbreed freely in nature & produce fertile offsprings.

d. are reproductively isolated from other sps. (Main basis of biological sps. concept.)





Genus



≡ Group of related Species.

≡ Genus has more characters in common in comparison to species of other genera.

Genus ①

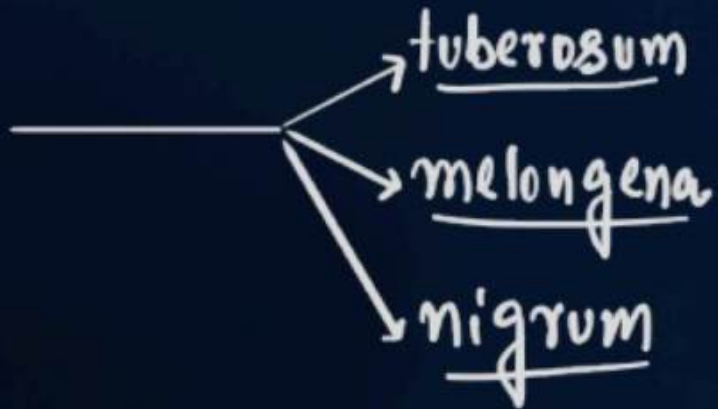
Sp_s A
Sp_s B
Sp_s C

Genus ②

Sp_s α
Sp_s β
Sp_s γ

≡ Each genus — may have one/more than one specific epithet.

Ex. 1. Solanum
(Genus)





FAMILY



≡ group of related genera.

≡ Plant families are characterised on the basis of

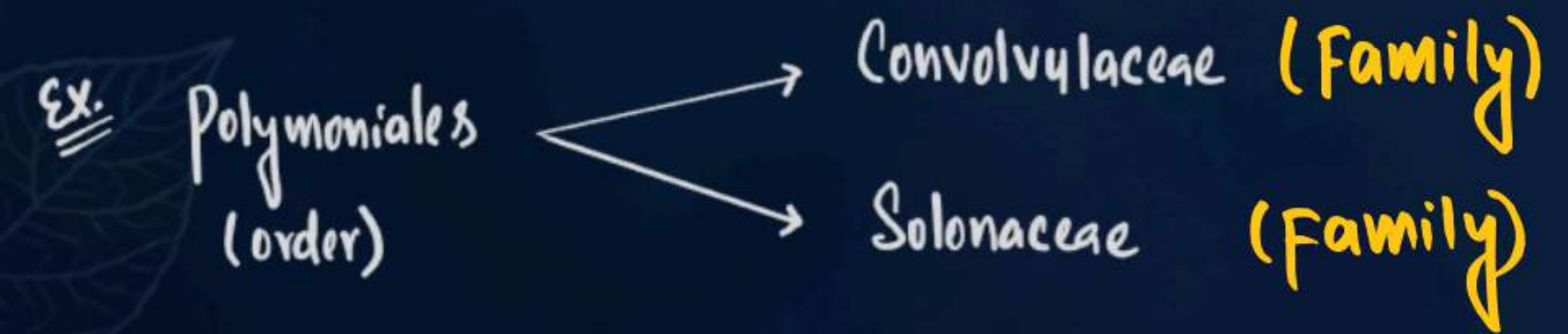
Vegetative characters ✓
AND
Reproductive characters. ✓





ORDER

≡ group of related Families.





CLASS



≡ group of related orders.

Dicots
↑

Ex. Dicotyledonae
(Class)

→ Sapindales (order)

→ Polymoniales (order)



Phylum/Division

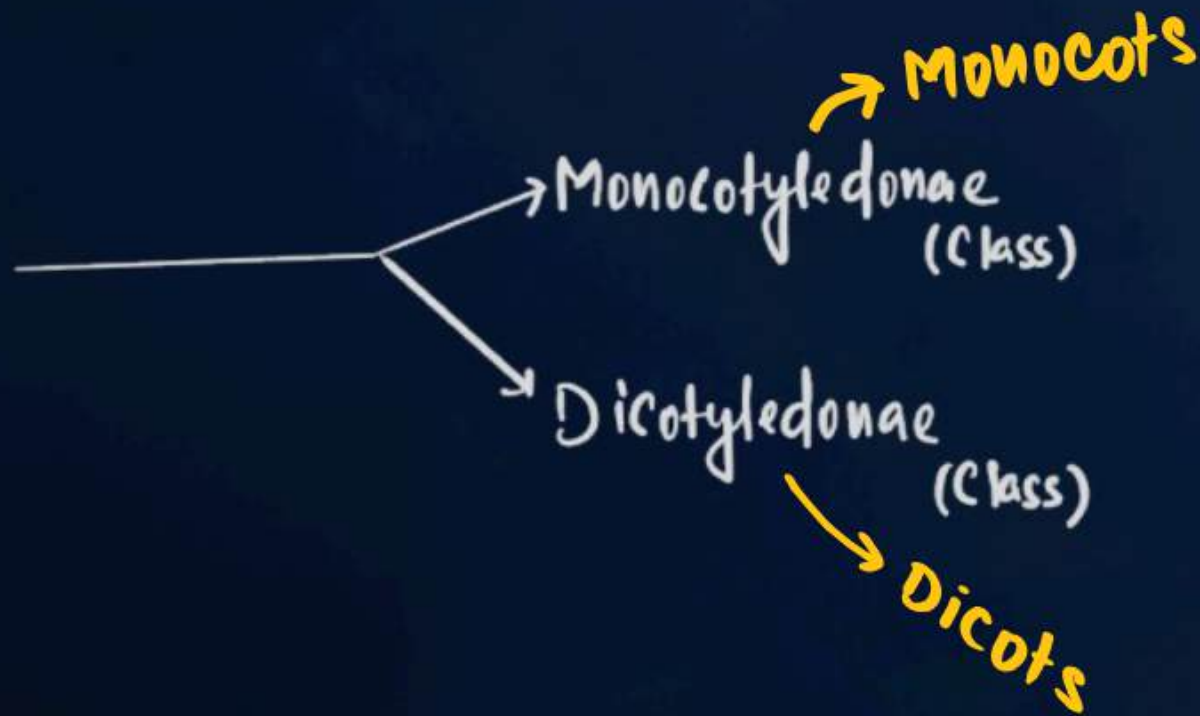


≡ Phylum — used in animal classification.
≡ Division — used in plant classification.

≡ Group of related classes.

Ex.

Angiospermae
(Division)





KINGDOM

≡ group of related Divisions / Phyla.

≡ highest taxonomic Category.

Example:

1. Plant Kingdom — includes All plants.

2. Animal Kingdom — includes All Animals.

NCERT

Common Name	Biological Name	Genus	Family	Order	Class	Phylum/ Division
Man	Homo sapiens	Homo	Hominidae	Primata	Mammalia	Chordata
Housefly	Musca domestica	Musca	Muscidae	Diptera	Insecta	Arthropoda
Mango	Mangifera indica	Mangifera	Anacardiaceae	Sapindales	Dicotyledonae	Angiospermae
Wheat	Triticum aestivum	Triticum	Poaceae	Poales	Monocotyledonae	Angiospermae

Herbarium



Home work



**IMPORTANT Questions
for JKBOSE 2026.**



1. Define:

1. Taxon ✓
2. Rank ✓
3. Taxonomy ✓
4. Systematics ✓
5. Species ✓
6. Biodiversity ✓



2. Define binomial nomenclature. write its rules ? *imp.*



3. Discuss Taxonomic hierarchy.

1. Define Herbarium.



Thank You

